Loop Engineering

General Agents, Chapter 8/12 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/loop-engineering

EN: From prompting to looping: a small system that starts itself, finds work, does it, gets it checked, updates memory, and calls a human only for decisions that matter. 2 things always stay human: intent and accountability.
Roman Urdu: Prompting se looping tak: chhota system jo khud shuru hota, kaam dhoondta, karta, check karwata, memory update karta, aur sirf zaroori decisions ke liye insaan bulata. 2 cheezein hamesha insaan ki: intent aur accountability.

1. Six Parts of a Loop + Heartbeats

- **Loop's 6 parts:** heartbeat, worktree, skill, subagents (maker-checker), connector (MCP), spine (state/memory -- 'no spine, no loop').
- **4 heartbeat kinds:** in-session, conditional/run-until-done, scheduled, event-driven. 3 required stops for any conditional loop: success condition, limit, no-progress check.

2. Body + Spine

- Worktrees keep parallel agents from overwriting each other's files. Skills put knowledge in one place since every run is a fresh session with no memory.
- **Maker-checker:** whoever builds something doesn't approve it themselves -- a separate (often cheaper) checker model reviews.
- **Spine = memory between runs** (rules file + progress.md). No spine, no loop -- delete progress.md and the loop relearns everything from scratch every time.

3. Dreaming + Human Control

- Dreaming (managed loop that proposes rule improvements from evidence) must NEVER run without a human gate -- defends against memory poisoning and brevity bias.
- **In/On/Out of the loop:** in (prompting), on (loop engineering, attention at gates), out (never acceptable -- happens by drift, not intention).
- 'Done' is a claim, not proof -- checking work is still always yours.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Loop's 6 parts	Heartbeat, worktree, skill, subagents (maker-checker), connector, spine.	Heartbeat, worktree, skill, subagents (maker-checker), connector, spine.
Spine	Memory between runs (rules file + progress.md) -- 'no spine, no loop'.	Runs ke beech ki memory -- 'spine nahi, loop nahi'.
Maker-checker	The thing that builds something never approves itself.	Jo banata hai wo khud approve nahi karta.
Dreaming	A managed loop proposing rule improvements -- always needs a human gate.	Managed loop jo rule improvements propose karta -- hamesha human gate chahiye.
On-the-loop	The system runs; your attention is only at the gates.	System chalta; aapka attention sirf gates par.

Quick Self-Check

Question	Answer
What are the two things that always stay human in a loop?	Intent (saying clearly what you want) and accountability (owning what ships).
What happens to a loop if you delete its progress.md file?	It loses its spine -- the loop relearns everything from day one, every single time ('no spine, no loop').
Why must a maker never approve its own work?	To avoid the same blind spots re-approving their own mistake -- a separate checker model catches what the maker can't see in itself.
Why must a 'dreaming' loop never run without a human gate?	It proposes changes to the system's own rules from evidence -- unreviewed, it risks memory poisoning and brevity bias (over-compressed, lossy rewrites).

Where This Leads / Sources

Leads to: Harness Engineering, Graph Engineering, Trusting the Checker. Full 12 practice projects + Routines appendix in 07-09; 61-Q quiz in 15-test-your-understanding.

Source: docs/loop-engineering/ (this repo), content originally via Zia Tutor AI connector.